

Extra Hidden States as Kullback-Leibler Corrections under Emission Misspecification in Hidden Markov Models

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Abstract

Hidden states in fitted hidden Markov models are often interpreted as latent regimes. This interpretation can fail under emission misspecification. We formalize a sufficient mechanism under which an additional fitted state is a Kullback-Leibler correction to a restrictive emission family, not an additional scientific regime. The main result compares observed laws only: for a persistent aggregate HMM whose emissions are identifiable finite mixtures from a dominated family, a refined HMM reproduces the observed process exactly, whereas every non-refined HMM with one fitted-family emission per aggregate regime has strictly positive relative-entropy rate. A Gaussian simulation and an enriched-emission refit illustrate the diagnostic implication. A three-state Gaussian HMM splits a heterogeneous aggregate regime, while a two-state HMM with two-component Gaussian emissions absorbs the same structure inside the emission distribution. Thus the fitted number of states can measure emission approximation order rather than the number of substantive regimes.

Keywords: hidden Markov model; misspecification; finite mixtures; relative entropy rate; Kullback-Leibler projection; state splitting.

1 Introduction

Hidden Markov models (HMMs) are widely used to segment dependent observations into latent states. In applications, these states are often interpreted as behaviours, regimes, or classes. This interpretation is attractive but fragile: a fitted state is simultaneously a scientific label and a component of a statistical approximation.

The fragility is most visible under emission misspecification. Suppose the data-generating process has a small number of aggregate regimes, but the conditional distribution within one regime is not contained in the emission family used for fitting. Then a likelihood-based HMM can create extra states to approximate the shape of that misspecified emission. A heavy-tailed, overdispersed, skewed, or heterogeneous regime may be split into several fitted states, even when there is no additional aggregate behaviour.

This empirical difficulty is well documented in HMM order selection. In particular, [Pohle et al. \[2017\]](#) show that standard information criteria can favour numbers of states that are difficult to interpret scientifically. One practical response is to enrich the observation model directly, for example with flexible nonparametric emissions as in [Langrock et al. \[2015\]](#). Likelihood theory for HMMs also shows that, under misspecification, maximum likelihood targets pseudo-true parameters minimizing limiting KL or relative-entropy criteria; see [Leroux \[1992\]](#), [Bickel et al. \[1998\]](#), [Douc and Moulines](#)

[2012]. General HMM inference is treated in Cappé et al. [2005] and Zucchini et al. [2016]. Our aim is to give a short structural mechanism explaining why a pseudo-true observed law can require more fitted states than the scientific number of regimes.

The main contribution is Theorem 5.1. It compares observed laws only. Under compactness, identifiable finite mixtures, pairwise distinct refined atoms, and uniformly positive aggregate transitions, a refined HMM reproduces the observed process exactly, whereas every non-refined HMM with the smaller number of states is bounded away from the truth in relative-entropy rate. The transition-positivity assumption excludes structural zero transitions; it is used as a transparent sufficient condition for a one-step predictive lower bound, not as a necessary condition. Two preliminary results prepare this theorem. Proposition 3.1 is an aligned comparison, meaning that the hidden aggregate trajectory is fixed. Proposition 4.1 records the classical state-space refinement behind mixture emissions, a construction closely related to standard HMM state augmentation [Rabiner, 1989, Zucchini et al., 2016].

The result is deliberately stated for a general dominated emission family. It applies whenever the finite-mixture geometry is identifiable and compact. Classical finite-mixture identifiability goes back to Teicher [1963] and Yakowitz and Spragins [1968]; HMM identifiability and order estimation are studied by Gassiat and Keribin [2000], Allman et al. [2009], Gassiat et al. [2016], and Lehéricy [2019]. The present paper uses mixture geometry directly through predictive densities. It should be read as a sufficient mechanism, not as a claim that every extra state is corrective.

2 Notation and emission-mixture geometry

Let $(Y, \mathcal{B}, \lambda)$ be a dominated observation space. For densities p, q , write

$$D(p\|q) = \int p(y) \log\{p(y)/q(y)\} \lambda(dy),$$

with the usual conventions. We use Pinsker's inequality,

$$D(p\|q) \geq \frac{1}{2} \|p - q\|_1^2. \quad (1)$$

Let $\mathcal{F} = \{f_\eta : \eta \in \Theta\}$ be the fitted emission family and define the class of mixtures of at most K fitted-family emissions by

$$\mathcal{M}_K = \left\{ \sum_{j=1}^K v_j f_{\eta_j} : v_j \geq 0, \sum_{j=1}^K v_j = 1, \eta_j \in \Theta \right\}. \quad (2)$$

Zero weights are allowed, so $\mathcal{M}_K \subseteq \mathcal{M}_{K+1}$.

For $K \geq 1$, let $\mathbf{H}_K(\mathcal{F})$ denote the class of observed laws generated by K -state HMMs with arbitrary initial distribution, arbitrary transition matrix, and emission densities in $\mathcal{F}$. Only the one-step predictive form of such laws is used below.

For stationary observed processes P, Q , define the lower relative-entropy rate

$$\underline{D}^{\text{rate}}(P\|Q) = \liminf_{T \rightarrow \infty} \frac{1}{T} D(P^{1:T} \| Q^{1:T}).$$

The liminf is enough for the lower bounds below.

Fix an integer $K_0 \geq 1$. In the applications below, K_0 is the number of aggregate scientific regimes.

Assumption 2.1 (Emission family). The parameter space Θ is compact. The map $\eta \mapsto f_\eta$ is continuous from Θ to $L^1(\lambda)$, and $f_\eta > 0$ λ -almost everywhere for every $\eta \in \Theta$.

Assumption 2.2 (Identifiable finite mixtures). Finite mixtures of $\mathcal{F}$ have identifiable minimal order. If two finite mixtures with strictly positive weights and pairwise distinct parameters are equal λ -almost everywhere, then they have the same number of components and agree up to label swapping.

Lemma 2.3 (Separation from lower-order mixtures). Assume Assumptions 2.1 and 2.2. Let $\eta_1^0, \dots, \eta_R^0$ be pairwise distinct, with $R > K_0$. For $c > 0$ such that $Rc \leq 1$, define

$$\mathcal{A}_c = \left\{ \sum_{r=1}^R v_r f_{\eta_r^0} : v_r \geq c, \sum_{r=1}^R v_r = 1 \right\}.$$

Then

$$d_c := \inf_{p \in \mathcal{A}_c} \inf_{q \in \mathcal{M}_{K_0}} \|p - q\|_1 > 0,$$

and hence

$$\inf_{p \in \mathcal{A}_c} \inf_{q \in \mathcal{M}_{K_0}} D(p \| q) \geq \frac{1}{2} d_c^2 > 0.$$

Proof. By Assumption 2.1, $\mathcal{M}_{K_0}$ is the continuous image of the compact set $\Delta_{K_0} \times \Theta^{K_0}$, hence compact in L^1 . The set $\mathcal{A}_c$ is also compact. The two sets are disjoint: if $p \in \mathcal{A}_c \cap \mathcal{M}_{K_0}$, then a mixture with exactly R positive-weight, pairwise distinct components equals a mixture whose zero-weight and repeated-atom components can be reduced to a minimal representation with at most K_0 components. This contradicts identifiable minimal order. Two disjoint compact sets in a metric space have positive distance. The KL bound follows from (1). $\square$

3 Aligned KL correction by state splitting

Let $Z_t \in \{1, \dots, K_0\}$ be a stationary aggregate Markov chain with transition matrix Γ^0 and stationary distribution π , with $\pi_k > 0$. Suppose

$$Y_t \mid Z_t = k \sim g_k.$$

For $m \geq 1$, define

$$\delta_k(m) = \inf_{q \in \mathcal{M}_m} D(g_k \| q).$$

Proposition 3.1 (Aligned KL improvement). If state k is represented by one emission from $\mathcal{F}$, the best aligned KL risk is

$$\mathcal{K}_{\text{unref}} = \sum_{k=1}^{K_0} \pi_k \delta_k(1).$$

If state k is refined into m_k substates with emissions from $\mathcal{F}$, the best aligned KL risk is

$$\mathcal{K}_{\text{ref}} = \sum_{k=1}^{K_0} \pi_k \delta_k(m_k).$$

Consequently, if $\delta_{k^*}(m_{k^*}) < \delta_{k^*}(1)$ for some k^* , then $\mathcal{K}_{\text{ref}} < \mathcal{K}_{\text{unref}}$.

Proof. In the aligned comparison, the hidden trajectory law is fixed at the true aggregate chain. Thus all transition terms cancel in the complete-data likelihood ratio. With one fitted emission f_{η_k} per aggregate state, stationarity gives risk

$$\sum_{k=1}^{K_0} \pi_k D(g_k \| f_{\eta_k}),$$

and taking the infimum over $\eta_1, \dots, \eta_{K_0}$ gives $\sum_k \pi_k \delta_k(1)$. With m_k substates inside aggregate state k , the aggregate fitted emission is any $q_k \in \mathcal{M}_{m_k}$; taking the infimum over these mixtures gives $\sum_k \pi_k \delta_k(m_k)$. Since $\mathcal{M}_1 \subseteq \mathcal{M}_{m_k}$, all terms weakly decrease, and one positive-weight strict decrease yields the claim. $\square$

4 Exact refinement with persistent aggregate dynamics

The next construction is the standard state-augmentation device used here to preserve the aggregate dynamics exactly.

Proposition 4.1 (Exact HMM refinement). *Suppose the aggregate emissions are*

$$g_k(y) = \sum_{a=1}^{m_k} w_{k,a} f_{\eta_{k,a}^0}(y), \quad w_{k,a} > 0, \quad \sum_a w_{k,a} = 1. \quad (3)$$

Let the refined states be (k, a) and define $r(k, a) = k$. Set

$$\Gamma_{(k,a),(\ell,b)}^{\text{ref}} = \gamma_{k\ell}^0 w_{\ell,b}, \quad (4)$$

and let $Y_t \mid X_t = (k, a) \sim f_{\eta_{k,a}^0}$. If X_1 has distribution $\pi_k w_{k,a}$ on (k, a) , then the observed law of the refined HMM is exactly the observed law of the aggregate HMM with transitions Γ^0 and emissions g_k .

Proof. The matrix in (4) is stochastic since $\sum_{\ell,b} \gamma_{k\ell}^0 w_{\ell,b} = 1$. The distribution $\pi_k w_{k,a}$ is stationary because

$$\sum_{k,a} \pi_k w_{k,a} \gamma_{k\ell}^0 w_{\ell,b} = w_{\ell,b} \sum_k \pi_k \gamma_{k\ell}^0 = \pi_\ell w_{\ell,b}.$$

Moreover,

$$\mathbb{P}\{r(X_t) = \ell \mid X_{t-1} = (k, a)\} = \sum_b \gamma_{k\ell}^0 w_{\ell,b} = \gamma_{k\ell}^0.$$

Finally, summing over all substate paths compatible with a fixed aggregate path $z_{1:T}$ gives

$$\pi_{z_1} \prod_{t=2}^T \gamma_{z_{t-1}, z_t}^0 \prod_{t=1}^T \sum_{a=1}^{m_{z_t}} w_{z_t, a} f_{\eta_{z_t, a}^0}(y_t) = \pi_{z_1} \prod_{t=2}^T \gamma_{z_{t-1}, z_t}^0 \prod_{t=1}^T g_{z_t}(y_t).$$

Summing over aggregate paths proves equality of all observed finite-dimensional laws. $\square$

5 Persistent marginal theorem

Let the aggregate transition matrix satisfy

$$\varepsilon := \min_{k,\ell} \gamma_{k\ell}^0 > 0. \quad (5)$$

This permits persistent regimes with large diagonal probabilities, while keeping every aggregate state present in every one-step prediction. Let

$$R = K_{\text{ref}} := \sum_{k=1}^{K_0} m_k, \quad w_{\min} := \min_{k,a} w_{k,a}, \quad c := \varepsilon w_{\min}.$$

Assume $R > K_0$ and the refined atoms $\eta_{k,a}^0$ are pairwise distinct. For this value of c , the condition $Rc \leq 1$ required in Lemma 2.3 is automatic, because the R refined predictive weights are all at least c and sum to one.

Theorem 5.1 (Persistent marginal KL separation). *Under Assumptions 2.1 and 2.2, the aggregate HMM with transitions Γ^0 satisfying (5) and emissions (3) satisfies*

$$\inf_{Q \in \mathcal{H}_{K_{\text{ref}}}(\mathcal{F})} \underline{D}^{\text{rate}}(P_0^Y \| Q) = 0,$$

but every K_0 -state HMM with emissions in $\mathcal{F}$ satisfies

$$\text{for every } Q \in \mathcal{H}_{K_0}(\mathcal{F}), \quad \underline{D}^{\text{rate}}(P_0^Y \| Q) \geq \frac{1}{2} d_c^2 > 0,$$

where d_c is the separation constant from Lemma 2.3. Hence

$$\inf_{Q \in \mathcal{H}_{K_{\text{ref}}}(\mathcal{F})} \underline{D}^{\text{rate}}(P_0^Y \| Q) < \inf_{Q \in \mathcal{H}_{K_0}(\mathcal{F})} \underline{D}^{\text{rate}}(P_0^Y \| Q).$$

Proof. The first claim follows from Proposition 4.1: the refined K_{ref} -state HMM reproduces the observed law exactly.

Now fix any candidate K_0 -state HMM with emissions $f_{\eta_1}, \dots, f_{\eta_{K_0}}$. Its one-step predictive density has the form

$$p_t^Q(y \mid y_{1:t-1}) = \sum_{j=1}^{K_0} \lambda_{t,j}(y_{1:t-1}) f_{\eta_j}(y),$$

so $p_t^Q(\cdot \mid y_{1:t-1}) \in \mathcal{M}_{K_0}$ for every history.

For the true aggregate HMM, let

$$\alpha_{t,k}(y_{1:t-1}) = \mathbb{P}_0(Z_t = k \mid Y_{1:t-1} = y_{1:t-1}).$$

Stationarity and (5) imply $\alpha_{1,k} = \pi_k \geq \varepsilon$. For $t \geq 2$,

$$\alpha_{t,\ell}(y_{1:t-1}) = \sum_k \mathbb{P}_0(Z_{t-1} = k \mid Y_{1:t-1} = y_{1:t-1}) \gamma_{k\ell}^0 \geq \varepsilon.$$

Therefore the true predictive density is

$$p_t^0(y \mid Y_{1:t-1}) = \sum_{k,a} \alpha_{t,k}(Y_{1:t-1}) w_{k,a} f_{\eta_{k,a}}^0(y),$$

with every refined weight at least $c = \varepsilon w_{\min}$. Thus $p_t^0(\cdot \mid Y_{1:t-1}) \in \mathcal{A}_c$ almost surely. By Lemma 2.3,

$$D\{p_t^0(\cdot \mid Y_{1:t-1}) \| p_t^Q(\cdot \mid Y_{1:t-1})\} \geq \frac{1}{2} d_c^2$$

almost surely. If any conditional divergence is infinite, the desired bound is immediate. Otherwise, the chain rule for relative entropy yields

$$D(P_0^{Y_{1:T}} \| Q^{Y_{1:T}}) = \sum_{t=1}^T \mathbb{E}_0 \left[D\{p_t^0(\cdot \mid Y_{1:t-1}) \| p_t^Q(\cdot \mid Y_{1:t-1})\} \right] \geq T \frac{1}{2} d_c^2.$$

Divide by T and take the liminf. □

Corollary 5.2 (Enriched-emission diagnostic). *Under the assumptions of Theorem 5.1, suppose the fitted emission family is enlarged from $\mathcal{F}$ to $\mathcal{M}_M$ for some $M \geq \max_k m_k$. Then*

$$\inf_{Q \in \mathcal{H}_{K_0}(\mathcal{M}_M)} \underline{D}^{\text{rate}}(P_0^Y \| Q) = 0.$$

Thus the refined K_{ref} -state HMM with simple emissions and the aggregate K_0 -state HMM with enriched emissions can both reproduce the observed law exactly.

Table 1: Main numerical diagnostic, $T = 1500$. Lower BIC is better.

Specification	K	logLik	BIC	interpretation
Gaussian, one Gaussian/state	2	-2223.7	4498.5	merged regime
Gaussian, one Gaussian/state	3	-2014.9	4132.2	split regime
Mixture, two Gaussians/state	2	-2012.3	4119.7	emission split

Table 2: BIC selection frequencies over fixed-seed replications. Lower BIC is better; $\Delta_G = \text{BIC}_{G,3} - \text{BIC}_{G,2}$ and $\Delta_M = \text{BIC}_{M,2} - \text{BIC}_{G,3}$.

T	reps	mean Δ_G	$\Pr(\Delta_G < 0)$	$\Pr(\Delta_M < 0)$
500	200	-52.4	0.870	0.945
1500	200	-220.9	0.905	0.960
5000	200	-903.5	0.970	0.985

Proof. Each aggregate emission g_k in (3) belongs to $\mathcal{M}_M$ when $M \geq m_k$. The original aggregate chain with emissions g_k is therefore a K_0 -state HMM with emissions in $\mathcal{M}_M$, and it has observed law P_0^Y . $\square$

Remark 5.3 (On the transition positivity assumption). The assumption $\min_{k,\ell} \gamma_{k\ell}^0 > 0$ is sufficient, not necessary. It gives a short one-step predictive proof and an explicit lower bound. A more abstract replacement is possible: compactness of a regular K_0 -state class, lower semicontinuity of the entropy-rate map, observed-law identifiability, and non-representability of at least one aggregate emission by $\mathcal{F}$ imply a positive separation by contradiction. That route is less transparent and is not needed for the present sufficient mechanism.

6 Numerical illustration

We simulated a persistent two-regime aggregate HMM with

$$\Gamma^0 = \begin{pmatrix} 0.93 & 0.07 \\ 0.05 & 0.95 \end{pmatrix}.$$

The first aggregate regime had a two-component Gaussian mixture emission, $0.70N(-1.20, 0.55^2) + 0.30N(1.10, 0.45^2)$, and the second regime had emission $N(3.80, 0.65^2)$. We fitted a Gaussian-emission HMM with $K = 2$, a Gaussian-emission HMM with $K = 3$, and an enriched $K = 2$ HMM with two Gaussian components per state. The replication script repeats the comparison for $T \in \{500, 1500, 5000\}$ with 200 fixed-seed replications per sample size and 10 random EM starts per replicated fit.

For the reference series in Table 1, the two-state Gaussian model uses one broad Gaussian to represent the heterogeneous first regime. The three-state Gaussian model improves the fit by splitting this heterogeneous regime. The enriched two-state model absorbs the same correction inside the emission distribution. The additional Gaussian-HMM state is therefore an emission-correction substate in the sense of the theory.

The replicated comparison confirms the diagnostic pattern. Within the simple Gaussian family, the three-state HMM is favoured in 87.0%, 90.5%, and 97.0% of the replications for $T = 500, 1500$, and 5000, respectively, because it can split the heterogeneous first regime. Once two-component Gaussian emissions are allowed, the two-state enriched model has lower BIC than the three-state Gaussian model in 94.5%, 96.0%, and 98.5% of the replications, in line with Corollary 5.2.

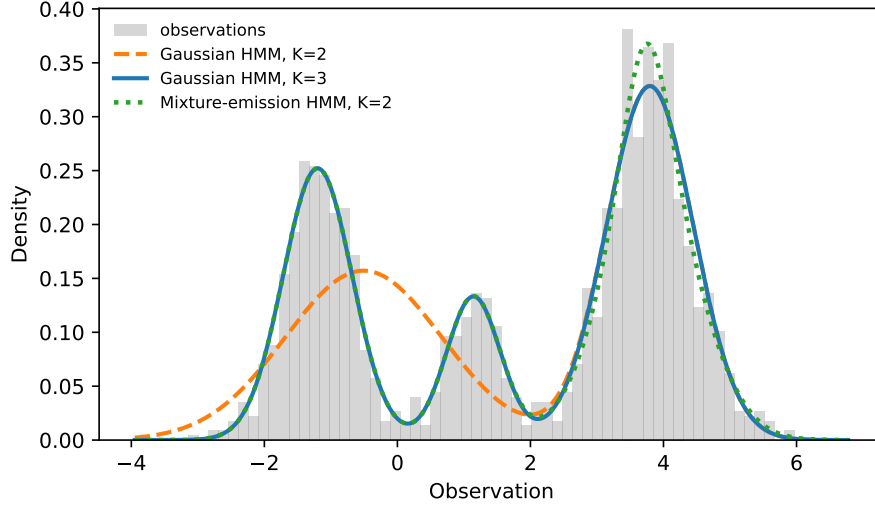


Figure 1: Histogram of observations and fitted marginal densities from Gaussian HMMs with $K = 2$ and $K = 3$ states. The three-state fit recovers two components inside the first aggregate regime.

7 Discussion

The results show that, under transparent sufficient conditions,

$$K_{\text{fitted}} = K_{\text{aggregate}} + \text{emission-correction substates}$$

can be the correct likelihood geometry. Extra states need not be spurious, but neither should they automatically be interpreted as new scientific regimes. They may be necessary to approximate a misspecified emission family.

The theory suggests a diagnostic workflow. After fitting a model with additional states, examine whether the extra state disappears when the emission family is enriched directly, for example by a heavy-tailed, overdispersed, copula, or explicit finite-mixture emission. If it does, the state was likely corrective. This distinction is relevant for likelihood-based state selection and for applied HMM interpretation, especially in domains where states are routinely assigned substantive meaning.

The exact theorem assumes that the aggregate emissions are finite mixtures from the fitted family. In applications, an emission may instead be heavy-tailed, skewed, or otherwise outside every finite mixture order considered in practice. The same interpretation remains useful but should be stated as approximation rather than exact representation: increasing the number of simple-emission states may successively reduce emission misspecification without producing a final finite-order equality. In that setting, the selected number of states can depend on sample size, the information criterion, and the richness of the competing emission family.

Data availability statement

The numerical code and generated outputs needed to reproduce the simulation are available at <https://github.com/AurelienNicosiaULaval/corrective-hmm-states>. No external data are used in the present draft.

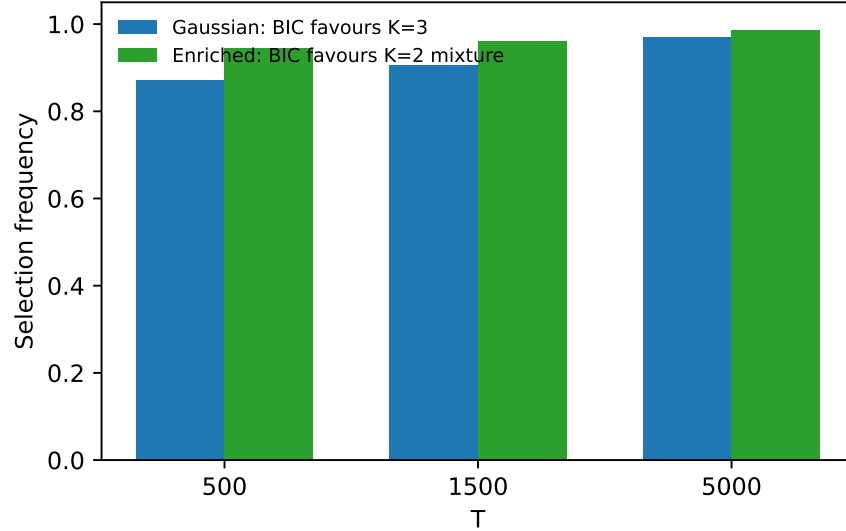


Figure 2: BIC selection frequencies across 200 replications for each sample size. The Gaussian comparison checks whether the simple-emission family favours $K = 3$ over $K = 2$; the enriched comparison checks whether a $K = 2$ mixture-emission HMM improves on the $K = 3$ Gaussian HMM.

Conflict of interest

The author declares no conflict of interest.

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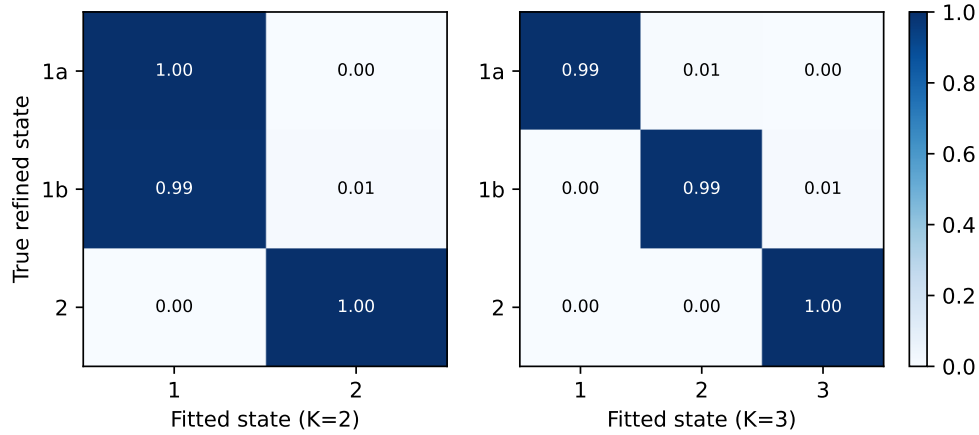


Figure 3: Viterbi decoding proportions for the reference series. The $K = 3$ Gaussian HMM separates the two refined substates inside the heterogeneous aggregate regime.

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Supporting information

The following supporting information is supplied separately: `supporting_information.pdf`, containing the mixture-emission EM details, supplementary BIC, posterior corrective-state, and residual ACF diagnostics, and a reproducibility note. The numerical files `review_replications.csv`, `review_summary_by_T.csv`, `review_model_comparison.csv`, and `review_metadata.json` accompany the supplement.